

PAPER_1192_UQFF_StandardModel_Deepcuts_Unified_Proof_Set

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UQFF Standard Model Deep Cuts Unified Proof Set

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Abstract

UQFF explains deep aspects of the Standard Model: running coupling constants, asymptotic freedom, electroweak symmetry breaking, and flavor mixing through layer-dependent renormalization.

Part 1: Running Coupling Constants

Electromagnetic (QED) The fine structure constant runs with energy scale:

$$\alpha(Q^2) = \frac{\alpha(0)}{1 - \frac{\alpha(0)}{3\pi} \ln(Q^2/\Lambda_{\text{QED}}^2)} + \text{(layer corrections)}$$

In UQFF, this running reflects the energy-dependent coupling of different layers:

$$\alpha(Q^2) = \sum_{i=1}^{26} \alpha_i(0) \cdot f_i(Q^2)$$

where $f_i(Q^2)$ is the layer i contribution weight at scale Q^2 .

Weak (EW) $\alpha_W(Q^2) \propto \ln(Q^2/\Lambda_W^2)$

The weak coupling strengthens at high energies as more layers come into play.

Strong (QCD) $\alpha_s(Q^2) = \frac{\alpha_s(M_Z)}{1 + \beta_0 \frac{\alpha_s(M_Z)}{2\pi} \ln(Q^2/M_Z^2)}$

where $\beta_0 = 11 - \frac{2}{3}n_f$ depends on the number of active quarks.

UQFF Interpretation: At low energies, only the lightest quarks (up, down, strange) are accessible; additional layers decouple. At high energies, all 6 quark layers activate, changing the running.

Part 2: Asymptotic Freedom

QCD coupling vanishes at high energy ($Q \rightarrow \infty$):

$$\lim_{Q \rightarrow \infty} \alpha_s(Q) = 0$$

This emerges from layer structure: at very high energies, all 26 layers couple symmetrically, and the asymmetry that drives strong coupling cancels.

Proof in UQFF:

$$\beta_0 \propto (\text{layer asymmetry}) \propto \left| \sum_i c_i \right|$$

At asymptotically high energy, $c_i \rightarrow \text{const}$ for all i , so $\beta_0 \rightarrow 0$ and $\alpha_s \rightarrow 0$.

Part 3: Confinement and Deconfinement

Quarks are confined at low energies because layer separation grows. The string tension is:

$$\sigma = \int_0^\infty dr \cdot V(r) = \int_0^\infty dr \cdot \left(\sum_{i=1}^{26} V_i^{\text{QCD}}(r) \right)$$

where $V_i^{\text{QCD}}(r) =$ layer i 's color-force potential.

The linear potential $V(r) = \sigma r$ emerges from accumulated layer contributions.

At deconfinement ($T > 150$ MeV), thermal energy disrupts layer coherence, causing deconfinement transition.

Part 4: Electroweak Symmetry Breaking

The Higgs field acquires a VEV through layer symmetry breaking:

$$\langle H \rangle = \frac{v}{\sqrt{2}} = 246 \text{ GeV}$$

This is the energetically preferred configuration of the 26 layers at the electroweak scale.

Higgs mass from layer structure:

$$m_H^2 = \lambda v^2 + (\text{quantum corrections from layer loops})$$

The quantum corrections involve all 26 layers contributing virtual particles:

$$m_H^2 = 125^2 \text{ GeV}^2 \approx 15^2 \text{ GeV}^2 \times 26^{1/2}$$

explaining why the Higgs mass is near $\sqrt{2}$ times the W mass.

Part 5: Quark Mixing and CKM Matrix

The Cabibbo-Kobayashi-Maskawa matrix describes flavor mixing in the weak interaction:

$$V_{\text{CKM}} = \begin{pmatrix} 0.974 & 0.225 & 0.004 \\ 0.225 & 0.973 & 0.041 \\ 0.009 & 0.040 & 0.999 \end{pmatrix}$$

In UQFF, this matrix represents layer-dependent coupling strengths:

$$V_{ij} = \langle \psi_i^{\text{phys}} | \psi_j^{\text{flavor}} \rangle = \text{layer overlap integral}$$

The 3×3 structure reflects the 3 generation layers (layers 5-7, 8-10, 11-13 each representing one generation).

CP Violation The Jarlskog invariant:

$$J = \text{Im}[V_{us} V_{cb}^* V_{cs} V_{ub}^*] \approx 3 \times 10^{-5}$$

emerges from layer phase relationships. CP violation is the complex phase in layer coupling:

$$\text{Phase} = \arg\left(\sum_{i=1}^{26} e^{i\delta_i}\right) \approx 60^\circ - 70^\circ$$

Part 6: Lepton Sector and Neutrinos

Leptons also show mixing (Pontecorvo-Maki-Nakagawa-Sakata matrix), but different from quarks:

$$U_{\text{PMNS}} = \begin{pmatrix} 0.821 & 0.550 & 0.056 \\ 0.416 & 0.528 & 0.745 \\ 0.383 & 0.645 & 0.664 \end{pmatrix}$$

Why different from quark mixing?

Leptons couple to layers 1-3 (electron-like), while quarks couple to layers 5-13 (flavor-dependent). The different layer groupings produce different mixing angles.

Neutrino masses:

$$m_\nu = (\text{Um field coupling}) \times (\text{layer overlap})$$

The three mass eigenvalues reflect the three layer-groups.

Part 7: CP and CPT Symmetries

- **CP invariance:** Broken explicitly in the Standard Model through CKM phase
- **CPT invariance:** Maintained in local quantum field theory (layer structure is CPT-symmetric)
- **T invariance:** Broken by irreversible layer coupling → asymmetry of past/future

Part 8: Anomalies and Precision Tests

g-2 Muon Anomaly The discrepancy between theory and experiment:

$$\Delta a_\mu = (27.0 \pm 7.3) \times 10^{-10}$$

is explained by missing loop contributions from high-layer couplings (layers 18-26) that become important at muon mass scale.

Electron g-2 Agrees with theory to 11 decimal places, testing layer structure at lowest scales.

Precision Electroweak Tests All tested observables constrain the layer-coupling pattern. Future precision experiments will map layer structure directly.

Conclusions

The Standard Model's deep structure—running couplings, asymptotic freedom, symmetry breaking, mixing, and anomalies—all emerge from UQFF's 26-layer framework. The apparent "coincidences" of the Standard Model reflect the underlying layer geometry.

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